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Roorkee, Rehmadpur, Uttarakhand 247667

Lab Sheet-04

Python Programming Lab – List and Dictionary Operations

MCA I Semester (Session 2026-2027)

Objectives:

- To practice fundamental operations on lists including searching, sorting, and manipulation.
- To implement dictionary-based programs for data storage, retrieval, and frequency analysis.

Requirements:

- Python 3.x installed on the system (IDLE, PyCharm, VS Code, or any preferred IDE).
- Understanding of Python lists and dictionaries.
- Knowledge of loops, conditionals, and input/output operations.

Program List:

1. Find the sum and average of elements in a list.
2. Rotate a list by n positions.
3. Remove all negative numbers from a list.
4. Find the product of all elements in a list.
5. Create a list of squares of numbers from 1 to n.
6. Split a list into two halves.
7. Check if a list is sorted or not.
8. Find pairs of numbers in a list whose sum is equal to a given number.
9. Flatten a nested list.

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10. Convert a list into a string without using `join()`.
11. Create a dictionary of student names and marks and display it.
12. Find the maximum and minimum marks from a dictionary.
13. Count the frequency of characters in a string using a dictionary.
14. Merge two dictionaries into one.
15. Sort a dictionary by keys.
16. Sort a dictionary by values.
17. Update the value of a key in a dictionary.
18. Remove a key from a dictionary.
19. Check if a key exists in a dictionary.
20. Find the sum of all values in a dictionary.
21. Find the key with the maximum value in a dictionary.
22. Create a dictionary with numbers as keys and their squares as values.
23. Convert two lists into a dictionary.
24. Invert a dictionary (keys become values and values become keys).
25. Count word frequency in a given paragraph using a dictionary.
26. Check if two dictionaries are equal.
27. Remove duplicate values from a dictionary.
28. Create a nested dictionary for student details (Name, Roll, Marks).
29. Find common keys between two dictionaries.
30. Create a dictionary from a string with characters as keys and frequency as values.

Instruction:

- Save each program in a separate .py file (e.g., Q11_student_marks.py).
- Use dynamic input wherever required (do not hardcode values).
- Add comments explaining your logic.
- Display outputs in a clear and formatted manner.

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- Ensure your code runs without syntax or runtime errors.

Guideline:

- Avoid using built-in shortcuts (like max(), min(), set(), etc.) unless specified.
- Use meaningful variable names.
- Follow Python indentation (4 spaces) strictly.
- Validate inputs before processing.
- Test each program with multiple cases**.**

Submission:

- Submit the source code files (.py) for each task along with a brief report documenting the implementation details and the results of the executions.



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